

An-Chi Liu

SENIOR SOFTWARE ENGINEER · AI AGENT RUNTIMES, LLM PLATFORMS & BACKEND INFRASTRUCTURE

Taiwanese, based in Tokyo, Japan (Permanent Resident)

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Summary

Builds agent runtimes and the platforms engineers adopt. Designed a Python **agent execution engine** with bounded-loop turns, schema-validated tool calls, and pluggable Claude, Codex, and OpenAI-compatible backends. Co-founded an **LLM/RAG product** with per-task model selection on cost, latency, and quality. Delivered T2's company-wide **evaluation platform**, a 12-engineer-month plan, **in 6 months**, creating **¥30M+** in value. Hands-on in Python, Go, and C++, with an **AI-driven workflow**: architecture by hand, implementation by coding agents, review and CI gating every merge.

Work Experience

T2, Inc. (L4 Autonomous Trucking)

Senior Software Engineer

Tokyo, Japan

Nov. 2025 – Present

- **Conceived, proposed, and delivered** a company-wide **evaluation and QA platform** from zero: design, budget, adoption. Shipped a **12-engineer-month plan in 6 months** over 7 major releases; 3,500+ jobs from 29 engineers, **2,100+ engineer-hours saved** (¥30M+ in direct and indirect value).
- Chose a **hybrid on-premises and cloud architecture** over pure cloud: on-site GPU servers for steady load, GCP VM workers for bursts. Built the full stack: Go backend, React frontend, containerized delivery.
- Built a **validation framework** scoring millions of runs against explicit metrics, wired into CI for every change. **Replaced thousands of engineer-hours** of manual review; evidence base for T2's **safety certification**.

Datra AI

Technical Co-founder

Remote

Feb. 2025 – Sep. 2025

- Co-founded an **LLM product** for Shopify merchants. Owned the architecture end to end (Go, React, PostgreSQL, AWS) and **managed two developers**.
- Designed the production **LLM/RAG pipeline** and **prompt engineering** for the application, including retrieval, summarization, and generation.
- **Per-task model selection**: evaluated OpenAI, Claude, and other models on cost, latency, and quality, and kept models swappable. Built a **human-in-the-loop evaluation pipeline** with metrics and benchmarks.

Mujin, Inc.

Backend Software Engineer

Tokyo, Japan

Sep. 2022 – May 2025

- Built **live streaming and playback for 20+ cameras under 50 ms** end-to-end latency with WebRTC, FFmpeg, and Go, chosen after benchmarking RTSP.
- Developed a **network discovery service** in C++ locating Ethernet/IP robots and cameras within seconds.
- Developed C++/Python **real-time robot control** and **warehouse systems**.

Open-Source Projects

solvcon

- Designed and built the **AI agent runtime** in Python: a **bounded-loop execution engine** with a step budget, where a failed tool call returns to the model as an observation to repair.
- **Pluggable model backends** behind one interface and registry: Claude, Codex, and OpenAI-compatible HTTP, switchable at runtime. **Guardrails**: sandboxed execution and JSON Schema validation of every tool call.
- **Context management**: state inventory and replayed history per request; cut tool definitions from **27 KB to 3.3 KB**.
- Wrote the project's **agent governance**: coding style, development guideline, and review process for coding agents.
- **Second-largest contributor (255 merged PRs, 67k lines)** to this C++/Python numerical toolkit: array core, 2D canvas, time-series and MCAP/ROS 2 ingestion, Pilot GUI. Presented at **PyCon JP 2024**. [PRs]

PcapPlusPlus

- **One of five core maintainers** since 2024 of a **3.1k-star** C++ packet-processing library: performance, CI/CD, protocol support.

Education

National Yang Ming Chiao Tung University (NYCU)

M.S. in Computer Science and Engineering

Hsinchu, Taiwan

Sep. 2019 – May. 2022

- GPA: 4.11/4.3. Thesis: **Offworker: An Offloading Framework for Parallel Web Applications**, published in **WISE 2022** (top-tier conference).
- **Best Master's Thesis Award**, Institute of Information & Computing Machinery, Taiwan (2022; one of 13 awardees).

National Taiwan University (NTU)

B.S. in Bio-Industrial Mechatronics Engineering

Taipei, Taiwan

Sep. 2014 – Jun. 2019

Skills

Programming	Proficient: C++11/14/17; Experience: Python 3, Go, TypeScript, JavaScript, C++20/23, Bash
AI / LLM	Agent runtimes, tool calling, guardrails, context management, RAG, pgvector, OpenAI/Anthropic APIs, coding agents
Cloud / Network	GCP, AWS, Docker, CI/CD, PostgreSQL, Linux, on-prem GPU clusters, ZMQ, WebRTC, Ethernet/IP, PcapPlusPlus, Git
Evaluation/Data	Evaluation pipelines, metrics and dashboards, regression testing, time-series processing, MCAP, ROS 2, pybind11
Languages	Mandarin Chinese (Native), English (Business), Japanese (Limited Conversational)